

Unit-4:Statistical Foundations for Data Science

Probability Distributions

A **probability distribution** describes how probabilities are assigned to different possible outcomes of a random variable.

A **random variable** is a variable whose value depends on chance.

Examples:

- Number of heads when tossing coins
- Marks scored in an exam
- Daily temperature
- Customer waiting time

Probability distributions are widely used in:

- Data Science
- Machine Learning
- Risk Analysis
- Business Forecasting

Discrete Random Variable

Definition:

A discrete random variable takes **countable values**.

Examples:

- Number of students in a class
- Number of defective products

Continuous Random Variable

Definition:

A continuous random variable takes **any value within a range**.

Examples:

- Height
- Weight
- Time

Types of Continuous Distribution: Uniform Distribution

In a uniform distribution, all values in a given interval are equally likely.

Probability Density Function

$$f(x) = \frac{1}{b - a}, \quad a \leq x \leq b$$

Characteristics

- Constant probability density
- Rectangular-shaped graph

Examples

- Random number generation
- Rolling a fair die (continuous approximation)

Normal Distribution

The normal distribution is the most widely used continuous distribution.

Probability Density Function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Characteristics

- Bell-shaped curve
- Symmetrical around the mean
- Mean = Median = Mode

Examples

- Human height
- Exam scores
- Measurement errors

Exponential Distribution

Used to model the time between events.

Probability Density Function

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Characteristics

- Right-skewed distribution
- Memoryless property

Examples

- Waiting time for a bus
- Time between customer arrivals

Gamma Distribution

A generalization of the exponential distribution.

Probability Density Function

$$f(x) = \frac{x^{k-1} e^{-x/\theta}}{\Gamma(k)\theta^k}$$

Characteristics

- Flexible shape
- Used for waiting-time models

Examples

- Rainfall modeling
- Insurance risk analysis

Beta Distribution

Defined on the interval $[0, 1]$.

Probability Density Function

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

Characteristics

- Flexible shape
- Suitable for probabilities and proportions

Examples

- Success probability estimation
- Bayesian statistics

Chi-Square Distribution

Derived from the sum of squares of standard normal variables.

Characteristics

- Positively skewed
- Depends on degrees of freedom

Applications

- Hypothesis testing
- Variance analysis

Student's t-Distribution

Used when sample size is small and population variance is unknown.

Characteristics

- Similar to normal distribution
- Heavier tails

Applications

- Small sample hypothesis testing
- Confidence intervals

F-Distribution

Used to compare variances.

Characteristics

- Right-skewed
- Depends on two degrees of freedom

Applications

- ANOVA (Analysis of Variance)
- Regression analysis

Types of Discrete Probability Distributions: Bernoulli Distribution

Used for a single trial with two outcomes: success/failure.

Random variable: $X \in \{0, 1\}$ Parameter: $p = P(\text{success})$

PMF:

$$P(X = 1) = p$$

$$P(X = 0) = 1 - p$$

Example: Tossing a coin once.

Binomial Distribution

Number of successes in n independent Bernoulli trials.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Parameters: n (trials), p (success probability) Mean: np Variance: $np(1 - p)$

Example: Number of heads in 10 coin tosses.

Poisson Distribution

Models number of events in a fixed interval of time/space.

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Parameter: λ (average rate) Mean = Variance = λ

Example: Number of calls per hour at a call center.

Geometric Distribution

Number of trials until first success.

Parameter: p

PMF:

$$P(X = k) = (1 - p)^{k-1}p$$

Example: Number of coin tosses until first head.

Negative Binomial Distribution

Number of trials until r -th success.

Parameters: r, p Generalizes geometric distribution

Example: Number of attempts until 3 successful shots.

Hypergeometric Distribution

Sampling without replacement.

Parameters: N (population), K (success items), n (sample size)

Example: Drawing cards from a deck without replacement.

Discrete Uniform Distribution

All outcomes are equally likely.

If outcomes = $1, 2, \dots, n$

$$P(X = x) = \frac{1}{n}$$

Example: Rolling a fair die.

Multinomial Distribution

Extension of binomial to more than two outcomes.

Used for categorical outcomes

Parameters: $n, p_1, p_2, \dots, p_k$

Example: Rolling a die many times and counting each face.

Sampling

Sampling is the process of selecting a subset (sample) from a population to make statistical inferences about the population.

- Population → Entire group of interest
- Sample → Part of population selected for study
- Census → Study of entire population

Why Sampling is Used

- Saves time
- Reduces cost
- Easier to handle than full population
- Useful when population is very large or infinite
- Sometimes destructive testing (bulbs, batteries)

Probability Sampling

Each unit has a known chance of selection.

- Simple Random Sampling (SRS)
- Stratified Sampling
- Systematic Sampling
- Cluster Sampling
- Multistage Sampling

Simple Random Sampling (SRS)

Every item has equal chance.

- Lottery method
- Random number table

Example: selecting 10 students from a class randomly.

Stratified Sampling

Population divided into strata (groups), then sample taken from each.

- Ensures representation from all groups

Example: dividing students by class (A, B, C) and sampling from each.

Systematic Sampling

Select every k -th element after a random start.

$$k = \frac{N}{n}$$

- Simple and quick method

Example: selecting every 5th person in a list.

Cluster Sampling

Population divided into clusters, then entire clusters are selected.

- Useful for large populations

Example: selecting whole schools instead of students.

Multistage Sampling

Sampling done in multiple steps.

- Country $\rightarrow$ District $\rightarrow$ School $\rightarrow$ Students

Non-Probability Sampling

Selection is not random.

- Convenience Sampling
- Judgmental/Purposive Sampling
- Quota Sampling
- Snowball Sampling

Non-Probability Sampling Types

- ① **Convenience Sampling:** Based on easy availability
- ② **Judgmental Sampling:** Based on researcher's judgment
- ③ **Quota Sampling:** Fixed number from categories, not random
- ④ **Snowball Sampling:** Participants refer new participants

Sampling Error and Bias

Sampling Error:

- Difference between sample and population value
- Always present
- Reduced by increasing sample size

Sampling Bias:

- Error due to improper selection

Causes:

- Non-random selection
- Undercoverage
- Voluntary response

Good Sample Characteristics

- Representative of population
- Adequate size
- Random selection
- Free from bias

Hypothesis

A hypothesis is a statement or assumption about a population parameter that can be tested statistically.

- Parameter: mean (μ), proportion (p), variance (σ^2)

Types of Hypothesis

Null Hypothesis (H_0)

- Represents no effect or no difference
- Assumed true unless evidence suggests otherwise

Example:

$$H_0 : \mu = 50$$

Alternative Hypothesis (H_1)

- Represents effect or difference exists

Example:

$$H_1 : \mu \neq 50$$

Types of Tests

- Two-tailed test: $H_1 : \mu \neq \mu_0$
- Left-tailed test: $H_1 : \mu < \mu_0$
- Right-tailed test: $H_1 : \mu > \mu_0$

Steps in Hypothesis Testing

- 1 State H_0 and H_1
- 2 Choose significance level (α)
- 3 Select test statistic
- 4 Compute test statistic
- 5 Find critical value / p-value
- 6 Make decision
- 7 Draw conclusion

Level of Significance

Probability of rejecting a true null hypothesis.

- Common values: 0.05 (5%), 0.01 (1%)

Errors in Hypothesis Testing

Type I Error:

- Rejecting H_0 when it is true
- Probability = α

Type II Error:

- Failing to reject H_0 when it is false
- Probability = β

Test Statistics

A value used to decide whether to reject H_0 .

- Z-test
- t-test
- Chi-square test
- F-test

Z-Test (Large Sample)

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

- Used when $n \geq 30$
- σ known

t-Test (Small Sample)

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

- Used when $n < 30$
- σ unknown

Decision Rule

- If test statistic $>$ critical value $\rightarrow$ Reject H_0
- If test statistic $\leq$ critical value $\rightarrow$ Fail to reject H_0

p-value Method

- If $p \leq \alpha \rightarrow$ Reject H_0
- If $p > \alpha \rightarrow$ Fail to reject H_0

Conclusion

- Reject $H_0 \rightarrow$ significant result
- Fail to reject $H_0 \rightarrow$ not significant result

Z-Test: Problem Statement

A sample of $n = 36$ students has mean score $\bar{X} = 52$. Population mean is $\mu = 50$, population standard deviation $\sigma = 6$. Test at 5% level of significance.

- Test: Two-tailed Z-test

Z-Test: Hypothesis

- $H_0 : \mu = 50$
- $H_1 : \mu \neq 50$

Z-Test: Calculation

Formula:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Substitute values:

$$Z = \frac{52 - 50}{6 / \sqrt{36}}$$

$$Z = \frac{2}{6/6} = \frac{2}{1} = 2$$

Z-Test: Decision

- Critical value at 5% (two-tailed): ± 1.96
- Calculated value: $Z = 2$

$$|Z| > 1.96$$

Decision: Reject H_0

Conclusion: Significant difference in mean

t-Test: Problem Statement

A sample of $n = 10$ students has:

- Mean = 28
- Sample SD = 5
- Population mean = 30

Test at 5% level of significance.

t-Test: Hypothesis

- $H_0 : \mu = 30$
- $H_1 : \mu \neq 30$

t-Test: Calculation

Formula:

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Substitute:

$$t = \frac{28 - 30}{5/\sqrt{10}}$$

$$t = \frac{-2}{1.58} \approx -1.27$$

t-Test: Decision

- $df = 9$, critical value = ± 2.262
- Calculated value: $t = -1.27$

$$|t| < 2.262$$

Decision: Fail to reject H_0

Conclusion: No significant difference in mean

Chi-Square Test: Problem

A die is thrown 60 times:

Face	1	2	3	4	5	6
Observed	8	10	9	11	12	10

Test if die is fair.

Chi-Square: Hypothesis

- H_0 : Die is fair
- H_1 : Die is not fair

Chi-Square: Calculation

Expected frequency:

$$E = \frac{60}{6} = 10$$

Formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\chi^2 = 1.0$$

Chi-Square: Decision

- $df = 5$
- Critical value = 11.07

$$\chi^2 = 1.0 < 11.07$$

Decision: Fail to reject H_0

Conclusion: Die is fair

Correlation

Correlation measures the degree and direction of relationship between two variables.

- Shows how one variable changes when the other changes

Types of Correlation

- Positive Correlation: both variables move in same direction
- Negative Correlation: variables move in opposite direction
- Zero Correlation: no relationship

Example:

- Income $\uparrow \rightarrow$ Expenditure $\uparrow$ (Positive)
- Price $\uparrow \rightarrow$ Demand $\downarrow$ (Negative)

Degree of Correlation

- Perfect positive: $+1$
- Perfect negative: -1
- No correlation: 0
- Strong correlation: close to ± 1
- Weak correlation: close to 0

Scatter Diagram Method

- Graphical method of correlation
- Points are plotted on a graph
- Pattern shows relationship between variables

Karl Pearson Correlation Coefficient

Formula:

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

- Range: -1 to +1
- Measures linear correlation
- Most widely used method

Spearman Rank Correlation

Formula:

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

- Used for ranked data
- Non-parametric method
- Useful for ordinal data

Properties of Correlation

- Unit-free measure
- Symmetric: $r(x, y) = r(y, x)$
- Range between -1 and +1
- Does not imply causation

Correlation vs Causation

- Correlation does not imply causation

Example:

- Ice cream sales and drowning both increase in summer

Interpretation of r

- 0 to $\pm 0.3 \rightarrow$ Weak correlation
- ± 0.3 to $\pm 0.7 \rightarrow$ Moderate correlation
- ± 0.7 to $\pm 1 \rightarrow$ Strong correlation

Uses of Correlation

- Economics (price-demand analysis)
- Business forecasting
- Science and research
- Social studies

Karl Pearson Correlation: Problem

Find Karl Pearson correlation coefficient for:

X	1	2	3	4	5
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Y	2	4	5	4	5
---	---	---	---	---	---

Karl Pearson: Table

X	Y	X^2	Y^2	XY
1	2	1	4	2
2	4	4	16	8
3	5	9	25	15
4	4	16	16	16
5	5	25	25	25

Karl Pearson: Summations

- $\sum X = 15$
- $\sum Y = 20$
- $\sum X^2 = 55$
- $\sum Y^2 = 86$
- $\sum XY = 66$
- $n = 5$

Karl Pearson: Formula

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{(n \sum X^2 - (\sum X)^2)(n \sum Y^2 - (\sum Y)^2)}}$$

Karl Pearson: Calculation (1)

Numerator:

$$5(66) - (15)(20) = 330 - 300 = 30$$

Denominator:

$$(5 \cdot 55 - 225) = 275 - 225 = 50$$

$$(5 \cdot 86 - 400) = 430 - 400 = 30$$

Karl Pearson: Final Answer

$$r = \frac{30}{\sqrt{50 \cdot 30}} = \frac{30}{\sqrt{1500}}$$

$$r = \frac{30}{38.73} \approx 0.775$$

Conclusion: Moderate positive correlation

Spearman Rank: Problem

Find Spearman rank correlation:

X	10	20	30	40	50
Y	12	24	33	45	52

Spearman Rank: Ranks

X	Rank X	Y	Rank Y
10	1	12	1
20	2	24	2
30	3	33	3
40	4	45	4
50	5	52	5

Spearman Rank: Differences

Rank X	Rank Y	d	d^2
1	1	0	0
2	2	0	0
3	3	0	0
4	4	0	0
5	5	0	0

$$\sum d^2 = 0$$

Spearman Rank: Formula

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Spearman Rank: Final Answer

$$r_s = 1 - \frac{6(0)}{5(25 - 1)}$$

$$r_s = 1$$

Conclusion: Perfect positive correlation

Causation: Meaning

Causation means a cause-and-effect relationship between two variables, where one variable directly produces a change in another.

- Example: Smoking $\rightarrow$ Lung cancer (possible causal relationship)

Causation vs Correlation

Correlation:

- Shows relationship between variables
- Does NOT imply cause

Causation:

- One variable directly affects another
- Cause $\rightarrow$ Effect relationship exists

Important Idea

- Correlation does not imply causation

Example:

- Ice cream sales increase
- Drowning cases increase
- (But ice cream does not cause drowning)

Types of Causation

- Direct causation: one variable directly affects another
- Indirect (spurious): relationship due to third variable
- Bidirectional: both variables affect each other

Examples of Causation Types

Direct:

- Rain $\rightarrow$ Flood

Spurious:

- Ice cream sales and drowning (confounder: summer heat)

Bidirectional:

- Stress Sleep problems

Conditions for Causation

- Strong association exists
- Cause happens before effect
- No confounding variables
- Logical mechanism exists

Confounding Variable

A hidden variable that affects both variables and creates a false relationship.

- Example: Hot weather increases both ice cream sales and swimming accidents

Why Causation is Hard to Prove

- Many variables influence outcomes
- Ethical limitations in experiments
- Hidden confounders
- Observational data limitations

Key Difference

- Correlation shows relationship only
- Causation shows cause-effect
- Correlation does not imply causation

Conclusion

- Correlation is necessary but not sufficient for causation
- Causation requires stronger evidence than statistical association

Bayesian Thinking: Meaning

Bayesian thinking is a method of reasoning where we update our beliefs based on new evidence.

- Probability is treated as a degree of belief
- Beliefs are updated when new data arrives

Core Idea

- Start with an initial belief (prior)
- Observe new evidence
- Update belief using data
- Final belief is called posterior

Idea: Learn from evidence and revise belief

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- $P(A|B)$ = Posterior
- $P(A)$ = Prior
- $P(B|A)$ = Likelihood
- $P(B)$ = Evidence

Key Terms

Prior Probability:

- Initial belief before data
- $P(A)$

Likelihood:

- Probability of data given hypothesis
- $P(B|A)$

Posterior Probability:

- Updated belief after evidence
- $P(A|B)$

Evidence:

- Overall probability of data
- $P(B)$

Steps in Bayesian Thinking

- 1 Start with prior belief
- 2 Observe new data
- 3 Compute likelihood
- 4 Apply Bayes' theorem
- 5 Update belief (posterior)

Example: Medical Test

- Disease probability (prior) is low
- Test result comes positive
- Use Bayes' theorem to update probability

Conclusion: Final probability depends on test accuracy and prior probability

Advantages

- Updates knowledge with new data
- Works well with uncertainty
- Flexible and logical
- Used in AI and machine learning

Applications

- Medical diagnosis
- Spam email filtering
- Machine learning
- Decision making under uncertainty
- Risk analysis

Key Idea Summary

- Prior + Evidence $\rightarrow$ Posterior
- Beliefs are updated, not fixed

Bayesian vs Frequentist Thinking

Feature	Bayesian	Frequentist
Prior knowledge	Used	Not used
Probability meaning	Belief	Frequency
Parameters	Random	Fixed
Output	Posterior probability	p-value / CI
Update learning	Yes	No

Bayesian Numerical 1: Medical Test Problem

A disease affects 1% of a population.

Given:

- $P(D) = 0.01$
- $P(\bar{D}) = 0.99$
- $P(T|D) = 0.99$
- $P(T|\bar{D}) = 0.05$

Find $P(D|T)$ when test is positive.

Medical Problem: Formula

$$P(D|T) = \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|\bar{D})P(\bar{D})}$$

Medical Problem: Calculation

$$P(D|T) = \frac{0.99 \times 0.01}{(0.99 \times 0.01) + (0.05 \times 0.99)}$$

$$P(D|T) = \frac{0.0099}{0.0099 + 0.0495}$$

$$P(D|T) = \frac{0.0099}{0.0594}$$

$$P(D|T) \approx 0.1667$$

Medical Problem: Conclusion

- Even after a positive test, probability of disease is low
- Final answer: $P(D|T) = 0.1667$

Bayesian Numerical 2: Urn Problem

An urn has:

- Bag A: 2 white, 3 black balls
- Bag B: 4 white, 1 black ball

A bag is chosen randomly and a white ball is drawn.
Find probability that it came from Bag B.

Urn Problem: Given Data

- $P(A) = \frac{1}{2}, P(B) = \frac{1}{2}$
- $P(W|A) = \frac{2}{5}$
- $P(W|B) = \frac{4}{5}$

Urn Problem: Formula

$$P(B|W) = \frac{P(W|B)P(B)}{P(W|A)P(A) + P(W|B)P(B)}$$

Urn Problem: Calculation

$$P(B|W) = \frac{(4/5)(1/2)}{(2/5)(1/2) + (4/5)(1/2)}$$

$$P(B|W) = \frac{4/10}{2/10 + 4/10}$$

$$P(B|W) = \frac{4}{6}$$

$$P(B|W) = \frac{2}{3}$$

Urn Problem: Conclusion

- Probability that white ball came from Bag B is $\frac{2}{3}$
- Final Answer: $P(B|W) = \frac{2}{3}$

Thank You!